

Ellen Su

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Education

Ph.D. Student, Center for Data Science

New York University

- Research interests: representation learning, NLP, cognitive science
- Advisor: Todd Gureckis
- GPA: 4.0 / 4.0.

2024 - present

B.S., Computer Science

Princeton University

- Minors: Applied Math and Cognitive Science
- Advisor: Tom Griffiths
- Thesis: *Revealing the Priors of Deep Learning Models Through Iterated Learning*

2019 - 2023

Publications

Su, E., Ho, M.K., Gureckis, T. (2025). Integration of Language and Experience via the Instructed Bandit Task. *Proceedings of the 47th Annual Meeting of the Cognitive Science Society, CogSci 2025*.

Su, E., Vellore, A., Chang, A., Mura, R., Nelson, B., Kassianik, P., & Karbasi, A. (2024). Extracting Memorized Training Data via Decomposition. *ArXiv, abs/2409.12367*

Su, E., Arevalo, J., Carpenter, A., & Singh, S. (2024). MOTIVE: A Drug-Target Interaction Graph For Inductive Link Prediction. *Neural Information Processing Systems. (spotlight)*

Arevalo, J., Su, E., Ewald, J.D., van Dijk, R., Carpenter, A., & Singh, S. (2024). Evaluating batch correction methods for image-based cell profiling. *Nature Communications, 15*.

Research Experience

Robust Intelligence

San Francisco, CA

Research Intern, PI: Amin Karbasi

Summer 2024

- Developed a decomposition jailbreak to extract training data from production LLMs
- Wrote and published a first author preprint by the end of internship (August 2024)

Broad Institute of MIT and Harvard

Cambridge, MA

ML Research Associate, PIs: Anne Carpenter, Shantanu Singh

2023 - 2024

- Used graph neural networks to advance drug target interaction (DTI) discovery
- Produced a co-first author spotlight publication in NeurIPS (June 2024) and second author publication in Nature Communications (March 2024)

Princeton University

Princeton, NJ

Senior Thesis, PI: Tom Griffiths

2022 - 2023

- Used convolutional neural networks (CNNs) as learning agents in an iterated learning chain to identify inductive biases in the models
- Demonstrated that machine priors manifested in sequential decision making behavior
- Presented results at the Program in Applied and Computational Mathematics Symposium

Junior Paper, PI: Ben Raphael

2021 - 2022

- Developed to a machine learning algorithm that predicts copy number variation mutations from spatial transcriptomics data

Fellowships and Awards

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| 2024 | Spotlight, NeurIPS 2024 (top 2.08% of submissions)
<i>MOTIVE: A Drug-Target Interaction Graph For Inductive Link Prediction.</i> |
| 2023 | Princeton CS Outstanding Student Teaching Award (\$500) |
| 2022 | Princeton CS Grace Hopper Celebration Grant (\$2500) |
| 2021 | Princeton Undergraduate Research Summer Fellowship (\$5000) |

Work Experience

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| Robust Intelligence
<i>Machine Learning Engineer Intern</i> | San Francisco, CA
<i>Summer 2024</i> |
| <ul style="list-style-type: none">• Researched and proposed implementation plan for multilingual support in AI firewalls• Built a multilingual dataset of prompt injections; benchmarked and evaluated multilingual language models in terms of performance and latency | |
| J.P. Morgan Chase & Co
<i>AI & Data Science Analyst</i> | Chicago, IL
<i>Summer 2022</i> |
| <ul style="list-style-type: none">• Created a decision-making tool (with Python, SQL, Alteryx, and Tableau) for the email marketing team to optimize their campaigns | |

Teaching Experience

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| Princeton Tutoring
<i>Tutor</i> | Princeton, NJ
<i>2021 - 2024</i> |
| <ul style="list-style-type: none">• Taught math, physics, and CS to middle and high school and college students | |
| Princeton Computer Science
<i>COS495 Undergraduate Course Assistant</i> | Princeton, NJ
<i>Spring 2023</i> |
| <ul style="list-style-type: none">• Assisted course staff, graded homework assignments, and answered student questions on course structure, assignments, and deliverables for Web3 and Blockchain course | |

Service

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| Princeton First Aid and Rescue Squad
<i>Emergency Medical Technician</i> | Princeton, NJ
<i>2021 - 2023</i> |
| <ul style="list-style-type: none">• Served as a first responder; stabilized, medicated, and transported patients• Certified by National Registry of Emergency Medical Technicians (NREMT) | |

Skills and Languages

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| Interests | Machine learning, cognitive science, artificial intelligence |
| Languages (program) | Python (Pytorch, Tensorflow), Java, Javascript, C, SQL, CSS |
| Languages | English (native), Mandarin (native), Spanish (advanced) |
| Interests | Reading, running, cooking |